MARKING SCHEME

Study-Uni

AC4101 — Management Accounting: Performance and Decision Making

Summer 2024 · Paper 1 · Full Marking Scheme

Module Code	AC4101
Module Title	Management Accounting: Performance and Decision Making
Examination Session	Summer 2024
Paper Number	Paper 1
Duration of Paper	1.5 hours
Internal Examiner	Patricia Healy BBS (Hons), FCA, AITI, FCIS, PDELT
Head of School / Department	Professor Michelle Carr
External Examiner	Professor Martin Quinn

Question	Scenario	Marks
Question 1	Paint Works Ltd — Traditional Absorption Costing vs Activity Based Costing	50
Question 2	Body Wise Ltd — Variance Analysis & Economic Order Quantity	50
Question 3	OptiCare Ltd — NPV, Payback Period & Accounting Rate of Return	50

Instructions to Candidates (as stated on the paper): Answer ANY TWO questions. Show all workings clearly – marks will be awarded for workings.

Note on this marking scheme: Numerical/computational parts (variances, ABC and absorption costing, NPV/Payback/ARR, EOQ) are marked primarily on correct method and workings, applying the **own-figure rule** throughout – an early error carried through consistently does not cause repeated loss of marks, and full credit is given for correct method even where the final figure is wrong because of an earlier slip. Discursive/written parts (memo-style analysis, behavioural discussion, non-financial factors) are marked using qualitative bands assessing depth of analysis, application to the specific scenario, and clear structure. As this is a closed-book, time-pressured exam rather than an essay assignment, no academic citation or referencing is expected or required in any answer.

QUESTION 1

Paint Works Ltd

Total: 50 marks

This question compares Traditional Absorption Costing with Activity Based Costing (ABC) for a paint manufacturer with three product lines. It tests overhead absorption mechanics, ABC cost driver calculation, profit statement preparation under both systems, and discursive understanding of over-absorption and the benefits of ABC.

Parts in this Question

Part	Requirement	Marks
(a)	Prepare a Profit Statement using Traditional Absorption Costing	10
(b)	Calculate the cost driver rates under the proposed ABC method	8
(c)	Prepare a Profit Statement using Activity Based Costing	20
(d)	Advise on TWO negative consequences of over-absorbing overheads	6
(e)	Explain TWO benefits of an Activity Based Costing system	6

Question 1(a)

10 marks

Prepare and present a Profit Statement using Traditional Absorption Costing.

Full Model Answer

	Primer	Matte	Oil Based	Total
Units	12,000	14,000	16,000	–
Sales (€)	2,400,000	4,200,000	7,200,000	13,800,000
Direct materials (€)	960,000	1,330,000	4,000,000	6,290,000
Direct labour (€)	600,000	910,000	1,600,000	3,110,000
Production overhead absorbed (€)	91,800	172,900	244,800	509,500
Total cost (€)	**1,651,800**	**2,412,900**	**5,844,800**	**9,909,500**
Profit (€)	**748,200**	**1,787,100**	**1,355,200**	**3,890,500**

Overhead absorbed per unit – calculated as (machine hours per unit × €1.75 machine hour rate) + (direct labour hours per unit × €1.20 direct labour hour rate):

	Primer	Matte	Oil Based
Machine OH: hrs/unit × €1.75	3 × 1.75 = 5.25	5 × 1.75 = 8.75	6 × 1.75 = 10.50
Assembly OH: hrs/unit × €1.20	2 × 1.20 = 2.40	3 × 1.20 = 3.60	4 × 1.20 = 4.80
OH absorbed per unit	**€7.65**	**€12.35**	**€15.30**

Reconciliation note: Total overhead absorbed across all three products is €509,500, compared with the actual overhead pool of €798,000 (€250,000 + €320,000 + €38,000 + €100,000 + €90,000). This is an under-absorption of €288,500, which is expected given the pre-set absorption rates (€1.75 machine hour rate, €1.20 direct labour hour rate) are based on a wider activity base than just these three products. This under-absorption is directly relevant to part (d).

Mark Allocation

Marks	Criteria
9–10	Correct profit statement for all three products with accurate sales, direct materials, direct labour, and overhead absorbed figures; correct total profit of €3,890,500; clear layout with workings shown for overhead absorption rates per unit.
7–8	Correct method throughout with one minor computational error (e.g. one product's overhead absorption miscalculated); own-figure rule applied consistently thereafter; layout clear.
5–6	Correct overall structure and method but with errors in two or more areas, or overhead absorption rates applied to the wrong activity base (e.g. using direct labour hours for machine overhead); sales and direct cost figures generally correct.
2–4	Some correct elements (e.g. sales revenue correctly calculated) but overhead absorption method fundamentally misapplied or omitted; partial attempt at profit statement structure.
0–1	No meaningful attempt, or profit statement omitted/unintelligible.

Question 1(b)

8 marks

Calculate the cost drivers under the proposed Activity Based Costing method.

Full Model Answer

Each cost pool's driver rate is calculated by dividing the budgeted cost pool by the total quantity for the period for its corresponding cost driver:

Cost Pool	Cost (€)	Cost Driver	Quantity for Period	Rate
Machining services	250,000	Machine hours	200,000	€1.25 per machine hour
Assembly services	320,000	Direct labour hours	175,000	€1.8286 per DLH
Set-up costs	38,000	Number of set-ups	1,200	€31.67 per set-up
Order processing	100,000	Customer orders	35,000	€2.8571 per order
Purchasing	90,000	Supplier orders	12,000	€7.50 per order

Note: The quantity for period figures (e.g. 200,000 machine hours, 35,000 customer orders) exceed the sum of the three products' own usage figures given later in the question. This reflects total activity across all of Paint Works Ltd's operations, not just these three product lines – the correct approach is always to use the company-wide "quantity for period" as the denominator when calculating the rate, then apply that rate to each individual product's own usage in part (c).

Mark Allocation

Marks	Criteria
7–8	All five cost driver rates correctly calculated (cost pool ÷ quantity for period) with workings shown; correct identification of which driver applies to which pool.
5–6	Four of five rates correct, or correct method throughout with minor arithmetic slips; own-figure rule applied.
3–4	Correct method (cost ÷ driver quantity) demonstrated but applied to two or more incorrect pairings of cost pool and driver, or several computational errors.
1–2	Some understanding shown (e.g. correctly identifies need to divide cost by activity) but rates largely incorrect or driver quantities misapplied (e.g. using product-level figures instead of period totals).
0	No meaningful attempt.

Question 1(c)

20 marks

Using your results from (b), prepare a Profit Statement using Activity Based Costing.

Full Model Answer

Step 1: Total activity consumption by product

	Primer	Matte	Oil Based
Machine hours (units × hrs/unit)	12,000×3 = 36,000	14,000×5 = 70,000	16,000×6 = 96,000
Direct labour hours (units × hrs/unit)	12,000×2 = 24,000	14,000×3 = 42,000	16,000×4 = 64,000
Number of set-ups	450	280	350
Customer orders	7,000	8,500	15,000
Supplier orders	3,000	4,000	4,500

Step 2: Overhead cost by activity (quantity × rate from part (b))

	Primer (€)	Matte (€)	Oil Based (€)
Machining (€1.25/hr)	45,000	87,500	120,000
Assembly (€1.8286/hr)	43,886	76,800	117,029
Set-up (€31.67/set-up)	14,250	8,867	11,083
Order processing (€2.8571/order)	20,000	24,286	42,857
Purchasing (€7.50/order)	22,500	30,000	33,750
Total ABC overhead	**145,636**	**227,452**	**324,719**

Step 3: ABC Profit Statement

	Primer	Matte	Oil Based	Total
Sales (€)	2,400,000	4,200,000	7,200,000	13,800,000
Direct materials (€)	960,000	1,330,000	4,000,000	6,290,000
Direct labour (€)	600,000	910,000	1,600,000	3,110,000
ABC overhead (€)	145,636	227,452	324,719	697,807
Total cost (€)	**1,705,636**	**2,467,452**	**5,924,719**	**10,097,807**
Profit (€)	**694,364**	**1,732,548**	**1,275,281**	**3,702,193**

Reconciliation note: Total ABC overhead recovered (€697,807) is, by design, less than the full €798,000 pool – consistent with part (b)'s note that period quantities reflect company-wide activity, not just these three products. All three products show lower profit under ABC than under traditional absorption costing, reflecting that ABC recognises Oil Based paint in particular consumes a disproportionate share of set-up, ordering and purchasing activity relative to its direct labour hour usage, which the traditional method ignores.

Mark Allocation

Marks	Criteria
17–20	Full, correct ABC profit statement for all three products, with clear workings showing activity consumption, overhead cost by activity, and final profit figures; correct total profit of €3,702,193 (or correct own-figure based on part (b) rates).
13–16	Correct method throughout with minor errors in one or two activity cost calculations; clear workings; own-figure rule applied consistently from part (b).
9–12	Correct overall ABC structure (multiple cost pools applied via activity consumption) but with errors across three or more cost pools, or some activity consumption figures (Step 1) miscalculated.
4–8	Partial ABC approach attempted – e.g. only some cost pools allocated via activity drivers, others still absorbed on a single volume basis; sales and direct cost figures generally correct.
0–3	No meaningful ABC method applied, or profit statement omitted/unintelligible.

Question 1(d)

6 marks

Advise the management team of Paint Works Ltd of TWO potential negative consequences of over-absorbing overheads and explain how this situation could be avoided.

Full Model Answer

Overstated Product Costs and Mispriced Output

Over-absorption occurs when the predetermined overhead rate recovers more cost than was actually incurred. If Paint Works Ltd sets absorption rates based on budgeted activity that turns out higher than actual usage, each unit of Primer, Matte or Oil Based paint will carry an inflated overhead charge. This can lead to products being priced above competitive market levels, particularly damaging in the commercial paint segment where customers are price-sensitive, potentially causing lost sales volume and market share.

Distorted Profitability and Poor Decision-Making

Where overhead is over-absorbed, reported product profitability is artificially suppressed at the unit level even though the year-end adjustment restores overall profit. Management reviewing interim product-line profitability (e.g. deciding whether to discontinue Primer or expand Oil Based capacity) could make poor strategic decisions based on distorted in-year cost data, before the over-absorption is corrected through the income statement at year end.

How This Can Be Avoided

Paint Works Ltd should review and revise its predetermined overhead absorption rates more frequently (e.g. quarterly rather than annually) so they better reflect actual activity levels as the year progresses. Moving to Activity Based Costing, as proposed in this question, also reduces the risk of broad, volume-based over- or under-absorption by tying overhead recovery more closely to the actual activities (set-ups, orders, machining) driving the cost in each product line.

Mark Allocation

Marks	Criteria
5–6	Two distinct, well-explained negative consequences of over-absorption (e.g. overstated costs/mispricing, distorted decision-making) each properly applied to Paint Works Ltd's context, plus a clear, practical explanation of how the situation could be avoided.
3–4	Two negative consequences identified with reasonable explanation, but limited application to Paint Works Ltd specifically, or the "how to avoid" element is underdeveloped or generic.
2	Only one negative consequence properly explained, or both consequences identified but with minimal explanation/application; some attempt at an avoidance method.
1	Vague or generic reference to over-absorption being "bad for the business" without specific consequences identified.
0	No meaningful attempt.

Question 1(e)

6 marks

Explain TWO benefits to Paint Works Ltd of using an Activity Based Costing system.

Full Model Answer

More Accurate Product Costing and Pricing Decisions

As shown in part (c), Oil Based paint absorbs proportionally far more set-up, ordering and purchasing activity than its direct labour hour usage alone would suggest under traditional costing. ABC recognises this by tracing overhead to the activities that actually drive it, giving Paint Works Ltd a more accurate picture of true product cost. This supports better pricing and product mix decisions – for example, identifying that Oil Based paint may be less profitable than the traditional statement suggested, prompting a review of its pricing or production volume.

Improved Cost Control and Overhead Management

By breaking overheads into specific activity pools (machining, assembly, set-ups, order processing, purchasing) rather than one or two broad cost centres, ABC gives management visibility into what is actually driving each category of cost. This allows Paint Works Ltd's management to target cost-reduction efforts directly – for instance, if set-up costs are high relative to other paint manufacturers, management can investigate batch sizes or production scheduling to reduce the number of set-ups required, rather than only being able to manage overhead as one undifferentiated total.

Mark Allocation

Marks	Criteria
5–6	Two distinct, well-explained benefits of ABC (e.g. costing accuracy, cost control/visibility) each clearly applied to Paint Works Ltd's specific scenario, ideally with reference to the case data (e.g. Oil Based paint's activity intensity).
3–4	Two benefits identified with reasonable explanation, but generic rather than applied to Paint Works Ltd specifically.
2	Only one benefit properly explained, or both identified with minimal explanation.
1	Vague or textbook-only reference to ABC being "more accurate" without explanation of why or how.
0	No meaningful attempt.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2

Body Wise Ltd

Total: 50 marks

This question tests standard costing and variance analysis (materials, variable overhead, labour rate and efficiency) for an orthopaedic products manufacturer, alongside discursive understanding of market share, plus a full Economic Order Quantity (EOQ) treatment including the underlying model, its calculation, and the relevant cost categories.

Parts in this Question

Part	Requirement	Marks
1(a)	Calculate total direct materials cost variance, variable overhead expenditure variance, direct labour rate variance, direct labour efficiency variance	24
1(b)	Outline TWO advantages and TWO disadvantages of increased market share	8
2(a)	Explain the EOQ model, its assumptions and calculation	6
2(b)	Calculate the EOQ for the knee replacement part	6
2(c)	Outline THREE types of inventory cost relevant to EOQ	6

Question 2 · Part 1(a)

24 marks

Calculate for the year: (i) Total direct materials cost variance; (ii) Total variable production overhead expenditure variance; (iii) Direct labour rate variance; (iv) Direct labour efficiency variance.

Full Model Answer

Standard cost card (hip component): Direct materials €150 per unit; Direct labour 3 hrs @ €45 = €135 per unit; Production overhead 3 hrs @ €60 = €180 per unit. Actual production for the year: 3,800 units. Actual direct labour hours worked: 9,500.

(i) Total Direct Materials Cost Variance

Standard material cost for actual production	3,800 units × €150 = €570,000
Actual material cost	€550,000
Total Direct Materials Cost Variance	**€20,000 Favourable**

(ii) Total Variable Production Overhead Expenditure Variance

Standard overhead cost for actual hours worked	9,500 hrs × €60 = €570,000
Actual variable overhead cost	€690,000
Variable OH Expenditure Variance	**€120,000 Adverse**

(iii) Direct Labour Rate Variance

Standard cost of actual hours worked	9,500 hrs × €45 = €427,500
Actual direct labour cost	€450,000
Direct Labour Rate Variance	**€22,500 Adverse**

(iv) Direct Labour Efficiency Variance

Standard hours for actual production	3,800 units × 3 hrs = 11,400 hrs
Actual hours worked	9,500 hrs
Favourable hours saved	1,900 hrs
Direct Labour Efficiency Variance	**1,900 hrs × €45 = €85,500 Favourable**

Reconciliation check: Rate variance (€22,500 A) + Efficiency variance (€85,500 F) = €63,000 F net, which equals the total direct labour cost variance (standard cost for actual output of €513,000 less actual cost of €450,000 = €63,000 F). This confirms the sub-variances are correctly calculated and reconcile to the total.

Mark Allocation

Marks	Criteria
21–24	All four variances correctly calculated with correct F/U signs and full workings shown (standard cost for actual output/hours clearly identified before comparison to actual cost).
16–20	Three of four variances fully correct; own-figure rule applied where an earlier variance feeds into a later one; correct method demonstrated throughout.
10–15	Correct method (standard vs actual comparison) shown for all four variances but with arithmetic errors in two or more, or F/U signs incorrectly assigned in multiple variances.
4–9	Some variances attempted with broadly correct approach (e.g. correct standard cost card derived) but formulae for rate/efficiency or expenditure variances confused or misapplied.
0–3	No meaningful attempt, or fundamental misunderstanding of standard costing (e.g. comparing budget units to actual units rather than using flexed/actual hours).

Question 2 · Part 1(b)

8 marks

Outline and briefly explain TWO advantages and TWO disadvantages associated with increased market share in an organisation.

Full Model Answer

Advantage: Economies of Scale

As Body Wise Ltd grows its market share in orthopaedic products, higher production volumes allow fixed costs (such as production overheads and administrative costs) to be spread across more units, reducing average cost per unit. This can be reinvested into pricing competitiveness or improved margins.

Advantage: Greater Bargaining Power

Increased market share typically gives a company stronger negotiating leverage with suppliers (e.g. for raw materials used in the hip component) due to higher purchase volumes, and with distributors or healthcare providers, supporting more favourable contract terms and supply security.

Disadvantage: Risk of Regulatory or Anti-Trust Scrutiny

As Body Wise Ltd's share of the orthopaedic products market grows, particularly if it approaches a dominant position, the company may attract increased regulatory attention regarding competition law, pricing practices, or market dominance, which can bring compliance costs and reputational risk.

Disadvantage: Increased Operational Complexity and Quality Risk

Growing market share generally requires scaling up production and distribution quickly, which can strain quality control processes – a particular concern for Body Wise Ltd given it supplies healthcare settings where product reliability (e.g. of the hip component) is safety-critical. Rapid expansion can also increase organisational complexity and dilute management oversight.

Mark Allocation

Marks	Criteria
7–8	Two clearly explained advantages and two clearly explained disadvantages, each going beyond a one-line definition to briefly explain the mechanism/impact; ideally with some application to Body Wise Ltd's healthcare/orthopaedic context.
5–6	Two advantages and two disadvantages identified with reasonable explanation, but largely generic/textbook rather than applied to the scenario.
3–4	Only three of the four required points adequately explained, or all four identified but with minimal explanation (definitions only).
1–2	One or two points identified with very limited explanation.
0	No meaningful attempt.

Question 2 · Part 2(a)

6 marks

Explain the Economic Order Quantity (EOQ) model, outlining the assumptions on which it is based and how it is calculated.

Full Model Answer

What EOQ Represents

The Economic Order Quantity model identifies the order quantity that minimises the total cost of inventory – specifically, the combined cost of placing orders (ordering cost) and the cost of holding inventory in stock (holding/carrying cost). As order size increases, ordering costs fall (fewer orders needed per year) but holding costs rise (more average inventory held); EOQ identifies the quantity at which these two opposing costs are minimised in total.

Key Assumptions

The model assumes: demand for the inventory item is known with certainty and occurs at a constant, uniform rate throughout the year; the cost of placing an order is fixed regardless of order size; the cost of holding one unit in stock for the period is constant; lead time for delivery is known and constant (no stockouts); there are no bulk-purchase discounts available for larger orders (unless a separate discount model is applied); and the unit purchase price itself does not vary with order size.

How It Is Calculated

EOQ is calculated using the formula $EOQ = \sqrt{(2DCo \div Ch)}$, where D is the annual demand in units, Co is the cost of placing one order, and Ch is the cost of holding one unit in stock for the period (typically one year). At the EOQ point, total annual ordering cost (number of orders × cost per order) equals total annual holding cost (average inventory of EOQ÷2 × holding cost per unit), confirming the minimum total inventory cost point.

Mark Allocation

Marks	Criteria
5–6	Clear explanation of what EOQ achieves (minimising combined ordering and holding cost), correct statement of the EOQ formula, and at least three relevant assumptions correctly outlined.
3–4	Reasonable explanation of EOQ's purpose and formula, but assumptions underdeveloped (fewer than three, or stated without explanation).
2	Correct formula stated but limited explanation of what EOQ represents or why it matters; assumptions largely omitted.
1	Vague or partial reference to EOQ as an inventory concept without meaningful explanation or formula.
0	No meaningful attempt.

Question 2 · Part 2(b)

6 marks

Calculate the Economic Order Quantity (EOQ) for the knee replacement part.

Full Model Answer

Annual demand (D)	9,000 units
Cost per order (Co)	€150
Cost of holding 1 unit for 12 months (Ch)	€9.75

EOQ = $\sqrt{(2 \times D \times Co \div Ch)} = \sqrt{(2 \times 9{,}000 \times 150 \div 9.75)} = \sqrt{(2{,}700{,}000 \div 9.75)} = \sqrt{276{,}923.08}$

Economic Order Quantity	**526 units (526.23 rounded)**

Verification check: At EOQ of 526 units, orders per year = 9,000 ÷ 526 = 17.10, giving an annual ordering cost of €2,565.39 and an annual holding cost of (526÷2) × €9.75 = €2,565.39. These two costs being equal confirms 526 units is correctly the minimum total cost point, as expected at the true EOQ.

Mark Allocation

Marks	Criteria
5–6	Correct EOQ formula applied with all figures correctly substituted, full workings shown, and a clearly stated final answer of approximately 526 units (or correctly rounded).
3–4	Correct formula and method, but a minor arithmetic or rounding error in the final calculation.
2	Correct formula identified/stated but one or more inputs substituted incorrectly (e.g. using monthly rather than annual demand).
1	Attempt made but formula incorrect or fundamentally misapplied.
0	No meaningful attempt.

Question 2 · Part 2(c)

6 marks

Outline the THREE types of costs to consider when deciding on the level of inventory to hold, and explain their importance in calculating Economic Order Quantity (EOQ).

Full Model Answer

1. Ordering (Procurement) Costs

These are the costs incurred each time an order is placed – for Body Wise Ltd, the €150 cost per order for the knee replacement part, covering administration, processing, and delivery arrangement. The more frequently orders are placed (i.e. the smaller the order quantity), the higher total annual ordering costs become – this is the Co term in the EOQ formula and is one half of the trade-off EOQ seeks to balance.

2. Holding (Carrying) Costs

These are the costs of keeping inventory in stock, including storage space, insurance, obsolescence risk, and the opportunity cost of capital tied up in inventory – for Body Wise Ltd, estimated at €9.75 per unit per 12 months for the knee replacement part. Larger order quantities increase average inventory held and therefore increase total holding costs – this is the Ch term in the EOQ formula and is the opposing force to ordering cost.

3. Stock-Out (Shortage) Costs

These are the costs incurred when inventory runs out before the next order arrives, including lost sales, production downtime, expedited/emergency order costs, and reputational damage – particularly significant for Body Wise Ltd given its healthcare customers may depend on timely supply of orthopaedic components for scheduled procedures. While not directly part of the basic EOQ formula itself, stock-out costs are the key reason businesses hold safety stock above the EOQ-derived reorder point, and are an essential practical consideration alongside the EOQ calculation.

Mark Allocation

Marks	Criteria
5–6	All three cost types correctly identified (ordering, holding, stock-out/shortage) with clear explanation of each and its relevance/role in the EOQ trade-off.
3–4	Three cost types identified but with uneven explanation (e.g. one or two explained well, one only named), or only two costs explained in full depth.
2	Two cost types correctly identified and briefly explained.
1	Only one cost type correctly identified and explained, or all three named with no explanation.
0	No meaningful attempt.

QUESTION 3

OptiCare Ltd

Total: 50 marks

This question tests investment appraisal – Net Present Value, undiscounted Payback Period, and Accounting Rate of Return – for a contact lens provider considering a new online patient service. It also tests discursive understanding of long-term profitability factors beyond the explicit forecast period, and non-financial considerations relevant to the investment decision.

Parts in this Question

Part	Requirement	Marks
(a)	Calculate the Net Present Value of the proposal	15
(b)	Calculate the undiscounted Payback Period of the proposal	5
(c)	Calculate the Accounting Rate of Return of the proposal	5
(d)	Identify TWO factors relevant to long-term profitability after Year 4	10
(e)	Explain THREE non-financial factors relevant to the decision	15

Question 3(a)

15 marks

Calculate the Net Present Value of the proposal.

Important judgement note (own-figure rule applies): The €3,500 research cost incurred over the past year is a **sunk cost** and must be excluded from the NPV calculation, as it has already been spent regardless of whether the project proceeds. The royalty fee (€45/patient) and home-kit materials cost (€75/patient) are treated here as **recurring annual variable costs**, applied to each year's estimated new patients – consistent with how revenue (average income per patient) is also calculated on a recurring annual basis. A candidate who instead treated the royalty fee as a single Year 0 lump sum (based on a literal reading of "payable upon commencement") should be given full own-figure credit provided the treatment is applied consistently and clearly explained.

Full Model Answer

Step 1: Year 0 cash outflow

Up-front licence fee	€57,000
New IT equipment	€40,000
Total Year 0 outflow	**€97,000**

The €3,500 research cost is correctly **excluded** – it is a sunk cost already incurred regardless of the investment decision and is not an incremental cash flow.

Step 2: IT maintenance cost by year

Year	Monthly Cost	Annual Cost
1	€200	€2,400
2	€200	€2,400
3	€205 (200 × 1.025)	€2,460
4	€205 (no further increase)	€2,460

Step 3: Net cash flow by year

	Year 1	Year 2	Year 3	Year 4
New patients	300	330	350	380
Revenue (patients × income/patient)	88,500	105,600	122,500	152,000
Royalty fee (€45/patient)	(13,500)	(14,850)	(15,750)	(17,100)
Home-kit materials (€75/patient)	(22,500)	(24,750)	(26,250)	(28,500)
Part-time administrator	(25,000)	(25,000)	(25,000)	(25,000)
IT maintenance	(2,400)	(2,400)	(2,460)	(2,460)
Net cash flow	**25,100**	**38,600**	**53,040**	**78,940**

Step 4: Discount to present value at 9% cost of capital

Year	Net Cash Flow	Discount Factor	Present Value
1	25,100	0.917	23,016.70
2	38,600	0.842	32,501.20
3	53,040	0.772	40,946.88
4	78,940	0.708	55,889.52
Total PV of inflows			**152,354.30**
Less: Year 0 outflow			(97,000.00)
NET PRESENT VALUE			**€55,354.30**

Decision: As the NPV is positive (€55,354.30), the HomeCare proposal should be ACCEPTED on financial grounds, as it is expected to increase shareholder wealth at OptiCare Ltd's 9% cost of capital.

Mark Allocation

Marks	Criteria
13–15	Correct exclusion of sunk research cost; all annual cash flows correctly built up (revenue and all relevant costs, with IT maintenance escalation correctly applied from Year 3); correct discounting using the given factors; correct final NPV of €55,354.30 (or correct own-figure if a clearly stated alternative royalty treatment is used) with a stated accept/reject conclusion.
10–12	Correct method and structure throughout with one or two minor errors (e.g. IT maintenance escalation timing, small arithmetic slip); sunk cost correctly excluded; discounting correctly applied.
6–9	Correct overall NPV structure (Year 0 outflow vs discounted future net cash flows) but with errors in several cash flow components, or sunk cost incorrectly included, or discount factors misapplied.
2–5	Some relevant cash flows identified and an attempt at discounting made, but fundamental errors in cash flow build-up or NPV mechanics (e.g. not discounting at all, or comparing only totals without time-weighting).
0–1	No meaningful attempt.

Question 3(b)

5 marks

Calculate the undiscounted Payback Period of the proposal.

Full Model Answer

Year	Net Cash Flow	Cumulative Cash Flow
0	(97,000)	(97,000)
1	25,100	(71,900)
2	38,600	(33,300)
3	53,040	19,740
4	78,940	98,680

Payback occurs during Year 3, once cumulative cash flow turns positive. At the start of Year 3, €33,300 of the initial investment remains unrecovered (cumulative cash flow of –€33,300 at end of Year 2).

Amount to recover at start of Year 3	€33,300
Year 3 net cash flow	€53,040
Fraction of Year 3 required	33,300 ÷ 53,040 = 0.6278
PAYBACK PERIOD	**2 years + 0.63 years ≈ 2.63 years (approx. 2 years 7.5 months)**

Mark Allocation

Marks	Criteria
5	Correct cumulative cash flow table built (using undiscounted net cash flows), correct identification of the year payback falls within, correct fractional-year calculation, and correct final answer of approximately 2.63 years.
3–4	Correct method and cumulative cash flow approach, with a minor error in the fractional-year calculation or own-figure carried correctly from part (a) cash flows.
2	Correct year of payback identified from cumulative cash flows, but fractional-year calculation omitted or incorrect.
1	Some attempt at cumulative cash flows but the year of payback incorrectly identified.
0	No meaningful attempt.

Question 3(c)

5 marks

Calculate the Accounting Rate of Return of the proposal.

Full Model Answer

Total net cash flow, Years 1–4	25,100 + 38,600 + 53,040 + 78,940 = €195,680
Average annual net cash flow	195,680 ÷ 4 = €48,920
Initial investment	€97,000
ARR	**48,920 ÷ 97,000 × 100 = 50.43%**

Basis used: ARR is calculated as Average Annual Net Cash Flow ÷ Initial Investment, as no separate depreciation charge or average (mid-life) investment figure is provided in the question. Where a candidate instead applies an average investment basis (Initial Investment ÷ 2 as the denominator), this should be given full own-figure credit provided the method is clearly shown and consistently applied; this alternative basis would give ARR = 48,920 ÷ (97,000÷2) × 100 = 100.86%.

Mark Allocation

Marks	Criteria
5	Correct average annual cash flow calculated from part (a)/(b) figures, correctly divided by a clearly stated investment base (initial or average), with correct final ARR percentage and workings shown.
3–4	Correct method and formula applied, with a minor arithmetic error, or correct own-figure based on a different (but valid and clearly stated) investment base.
2	Correct formula identified (average profit/cash flow ÷ investment) but with an error in calculating the average annual cash flow or in selecting the investment base.
1	Some understanding of ARR shown but formula substantially misapplied.
0	No meaningful attempt.

Question 3(d)

10 marks

Identify TWO factors the company should consider when assessing the long-term profitability of the HomeCare proposal, that is, after the initial four years.

Full Model Answer

System Upgrade and Replacement Costs

The question states the HomeCare system is assumed not to require upgrading for 6 years, meaning Years 5 and 6 benefit from continued patient revenue without further major capital outlay – but OptiCare Ltd must plan for a replacement or upgrade decision at the end of Year 6, including whether a new licence fee, IT equipment, and royalty/contract terms would apply, and whether patient volumes and income levels (which were only forecast to Year 4) would continue to grow, plateau, or require renegotiation of the HomeCare contract terms.

Sustainability of Patient Growth and Market Saturation

New patient numbers grew from 300 in Year 1 to 380 in Year 4 – OptiCare Ltd needs to consider whether this growth trajectory is sustainable into Years 5 and 6, or whether the addressable market for online/home contact lens services will saturate, plateau, or even decline as competitors potentially introduce similar online offerings, which would directly affect whether the strong cash flows seen in Year 4 continue, stabilise, or erode in the unforecast later years.

(Additional acceptable point) Technology Obsolescence and Competitive Response

Patient-facing healthcare technology evolves quickly; competitors may introduce more advanced systems within the 6-year window, eroding HomeCare's competitive advantage and patient retention before OptiCare Ltd would otherwise need to upgrade, which could shorten the useful economic life below the 6 years assumed and affect long-term profitability projections.

Mark Allocation

Marks	Criteria
8–10	Two distinct, well-developed factors relevant specifically to the post-Year-4 period (not simply repeating the Year 1–4 cash flow drivers), each properly explained with clear reference to the 6-year assumed useful life and the scenario's specific data (e.g. patient growth trend, no upgrade needed for 6 years).
6–7	Two relevant factors identified with reasonable explanation, but more generic in nature or with limited application to OptiCare Ltd's specific situation/data.
4–5	One factor well explained and applied; second factor underdeveloped, vague, or not clearly distinguishable from the first.
1–3	Factors identified relate to the Year 1–4 forecast period rather than genuinely addressing "after the initial four years", or are extremely generic.
0	No meaningful attempt.

Question 3(e)

15 marks

Explain THREE non-financial factors that the company should consider prior to making their final decision in deciding whether to proceed with the HomeCare system.

Full Model Answer

1. Patient Data Privacy, Security and Regulatory Compliance

HomeCare involves patients uploading personal health information online from home, which raises significant data protection obligations (e.g. GDPR compliance in an Irish/EU healthcare context) and cybersecurity risk. OptiCare Ltd must consider whether adequate safeguards, consent processes, and breach-response procedures are in place before committing to a system handling sensitive patient information remotely, as a data breach could cause reputational damage and regulatory penalties well beyond the direct financial cost modelled in the NPV.

2. Quality of Patient Care and Clinical Risk

Moving from in-clinic to remote/home-based assessment for contact lens patients carries clinical risk if issues that would normally be identified during an in-person examination (e.g. early signs of eye infection or improper lens fit) are missed through a remote scanning process. OptiCare Ltd needs to assess whether the HomeCare technology can maintain an equivalent standard of clinical safety and patient outcomes compared to traditional in-clinic visits before proceeding.

3. Patient Acceptance, Digital Literacy and Customer Experience

The success of HomeCare depends on patients being willing and able to use the online/home-based system – not all patients, particularly older or less digitally confident customers, may be comfortable or capable of using the scanning equipment at home. OptiCare Ltd should consider patient demographics, the need for training or support materials, and the risk of alienating part of its existing customer base who may prefer to continue using traditional in-clinic services.

(Additional acceptable point) Staff Impact and Organisational Change

Introducing HomeCare changes the role of clinic-based staff and requires the new part-time administrator role to be properly integrated; OptiCare Ltd should consider staff training needs, potential resistance to changing working practices, and whether existing staff have the skills to support a more technology-driven service model.

Mark Allocation

Marks	Criteria
13–15	Three distinct, well-developed non-financial factors, each clearly explained with specific application to OptiCare Ltd's contact lens/healthcare context (not generic investment-appraisal boilerplate); demonstrates genuine understanding of the clinical, data-privacy, and patient-experience dimensions of this specific proposal.
10–12	Three relevant factors identified with reasonable explanation, but one or more are less developed or only loosely applied to the specific scenario.
6–9	Two factors well explained and applied; third factor vague, underdeveloped, or essentially a restatement of one of the other two.
3–5	One factor well explained; remaining factors only named without meaningful explanation, or all three factors are generic and not applied to OptiCare Ltd specifically.
0–2	No meaningful attempt, or factors listed are purely financial in nature (misreading the question's "non-financial" requirement).

QUESTION 3 TOTAL — 50 MARKS